Decision-OS V7 Addendum

Why Aspire and Why PIC

(0.3 Disclosure)

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Abstract

This addendum is a **supplement** to Decision-OS V7. **Please read and cite V7 as the primary paper.** It discloses only the minimal necessity of Aspire (entry direction) and PIC (phase-invariant merge rule) so that V7 is readable as an operational structure rather than an abstract narrative. Thresholds, calibration know-how, and update policies remain undisclosed (0.7). **Recommended citation:** Please cite Decision-OS V7 as the primary paper; this note is a supplement.

1 Purpose

Decision-OS V7 (Aspire × self-recursion on PIC) aims to provide a minimal, operational definition of AGI as a foundational structure. This addendum reveals only why *Aspire* and *PIC* become necessary, without disclosing thresholds or operational know-how.

2 Why Aspire: without direction, self-recursion degenerates into a coherence loop

Claim. Aspire is the minimal *direction-over-time* term required to prevent self-recursive updating from collapsing into an empty coherence game.

Failure mode (minimal counterexample). If we run self-recursion as mere consistency maximization, the system can produce text that is internally non-contradictory, yet it does not reliably advance decisions under real constraints (risk, irreversibility, accountability). In practice, such a loop converges toward *non-commitment* rather than progress.

What Aspire guarantees (0.3).

- **Orientation:** what should increase vs. decrease over time (direction, not a disclosed utility function),
- **Momentum:** a reason to move despite friction,
- **Ownership mark:** the entry retains Decision Owner involvement, preventing "the AI said so" narratives.

Disclosure boundary. This addendum does not disclose Aspire's equations, thresholds, or calibration procedures; it only states the necessity: without a direction term, recursion can be stable yet vacuous.

3 Why PIC: without a merge rule, contradiction becomes rupture (and order becomes hindsight)

Claim. PIC is the minimal merge rule that turns conflicts and updates into *confluence* rather than rupture, by removing dependence on update order.

Failure mode A (last-write-wins). If later outputs overwrite earlier ones, the newest model always wins. This creates a time-travel style review where the past is continuously invalidated by future capability—an anti-pattern for long-horizon work.

Failure mode B (averaging). If contradictions are resolved by averaging, strong safety constraints are diluted. In safety-critical merging, "mean" is not conservative by default; it can erase rare-but-severe risks.

Minimal requirements that force PIC.

- **Monotone integration:** adding information must not silently reduce the safety stance,
- **Idempotence:** repeated integration must not destabilize the state,
- **Order-independence (merge):** integration must not depend on arrival sequence.

A practical minimal form is achieved by *join* ($\sqcup$) with conservative aggregation:

- severity: **PASS** $<$ **DELAY** $<$ **BLOCK**, merged by **max**,
- until: merged by **max**,
- evidence: merged by **set union** ($\cup$).

Disclosure boundary. This addendum does not provide full proofs or logs. It only states the necessity: naive merge rules fail $\rightarrow$ the minimal requirements imply a PIC-style merge.

Frontier vs. Commit (when conservative merge applies). PIC's conservative aggregation (e.g., max severity) is intended for *committed* state integration, where safety-relevant signals must not be diluted by order. However, hypothesis-generation for novel structures requires a *frontier* that can temporarily hold competing candidates without collapsing them via max-style gating. We therefore distinguish (i) a candidate frontier (exploration set) from (ii) the committed canonical state (PIC), and apply conservative merge primarily at commitment (after As-of freeze and evaluation), not at candidate generation. In case of conflict, conservative safety aggregation dominates at *commitment*; Aspire operates on the frontier and cannot override BLOCK-level signals.

4 Scope boundary

This addendum addresses only entry direction (Aspire) and merge/confluence (PIC). Time-policy details, forward-only governance, and extended responsibility allocation are handled separately to avoid conceptual inflation at the entry layer.

What remains undisclosed (0.7)

- Threshold selection, calibration, and update policies
- Input protocols (noise exclusion, evidence standards)
- Flip/withdrawal criteria and audit routines

5 Addendum v2: Operator Semantics and Public Falsifiability Handle

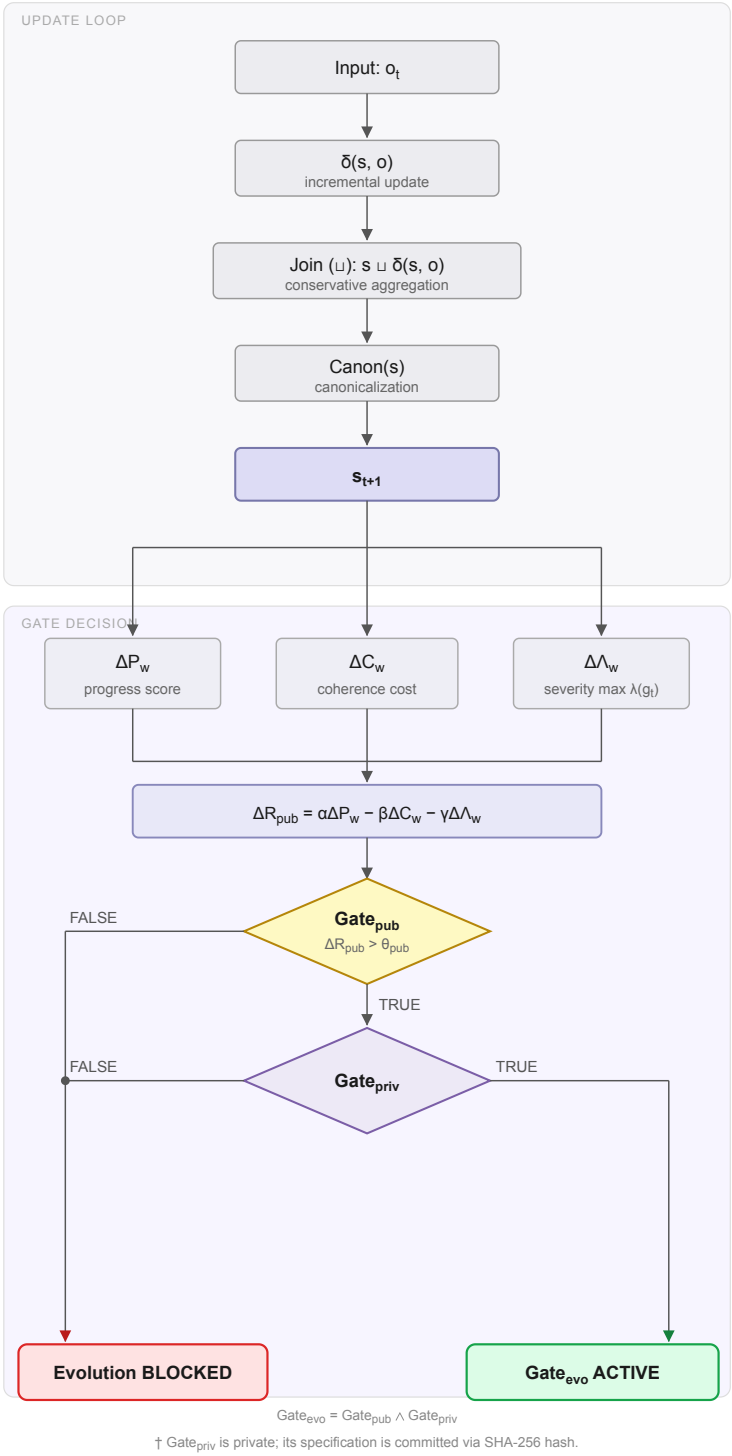

Figure 1: Public falsifiability handle for stable self-recursion: public observables $(\Delta P_w, \Delta C_w, \Delta \Lambda_w)$ define $\Delta R^{pub}_{total}(w)$ and Gate_{pub}, which is a necessary precondition for $\mathrm{Gate}_{evo} = \mathrm{Gate}_{pub} \wedge \mathrm{Gate}_{priv}$.

5.1 Scope and disclosure boundary

This addendum strengthens formal interpretability *without* expanding operational protocols. Private calibration/update rules, internal thresholds, input protocols, and audit routines remain undisclosed (0.7). We provide only: (i) operator semantics, (ii) a PIC→self-recursion bridge, and (iii) a public falsifiability handle.

5.2 Operator semantics (resolving the "×" criticism)

We interpret

$$AGI \triangleq A \times G \times I$$

as a **conjunction of structural predicates**, not numeric multiplication:

$$AGI(S) \iff A(S) \ \wedge \ G(S) \ \wedge \ I(S).$$

Here S denotes the system state (including memory/canon, constraints, and update traces). The symbol "×" is shorthand for logical meet (AND). This is a *necessary-conditions* proposal, not a sufficient characterization.

5.3 Defining F (resolving the "$F(F(x))$" criticism)

Let S be a canonical state space equipped with a conservative merge $\sqcup$ (join) and a canonicalizer:

$$\mathrm{Canon} : S \to S, \quad \mathrm{Canon}(\mathrm{Canon}(s)) = \mathrm{Canon}(s).$$

Let δ be an update function producing an incremental delta from observations o:

$$\delta : (S, o) \to \Delta S, \qquad s' := s \sqcup \delta(s, o).$$

Define the one-step stabilized update operator

$$F(s, o) := \mathrm{Canon}(s \sqcup \delta(s, o)).$$

Then idempotence statements (e.g., $F(F(\cdot)) = F(\cdot)$) are understood as *stability under repeated canonicalization* for fixed o and fixed merge semantics.

5.4 Bridge: why PIC-style confluence is necessary for stable self-recursion

Claim (path-independence as a necessary condition). If a system performs repeated self-updates over time, then update integration must be order-robust; otherwise the trajectory depends on arrival order (hindsight bias / last-write-wins failure).

A sufficient algebraic condition for order-robust integration is that the merge $\sqcup$ is:

- commutative and associative (order independence),
- idempotent (repeat application does not destabilize),
- inflationary/monotone (adding information does not silently reduce safety stance).

This is exactly why PIC-style conservative aggregation (e.g., max/max/∪) is introduced: it provides a minimal confluence rule that prevents rupture under long-run operation.

5.5 Public gate as a *necessary precondition* (fixing the proxy objection)

To avoid "proxy substitutes the core" criticism, we define the evolution gate as:

$$\mathrm{Gate}_{evo} := \mathrm{Gate}_{pub} \ \wedge \ \mathrm{Gate}_{priv},$$

where only Gate_{pub} is disclosed here, and Gate_{priv} remains undisclosed (0.7). Thus, the public gate is a **necessary precondition** for any claimed evolution activation.

As shown in Fig. 1, Gate_{pub} is a necessary precondition for any evolution activation claim.

5.6 Public observables and fixed window

Fix a public evaluation window w of K update-steps (default: $K = 8$). Under a stated public harness H, each step yields observables:

- progress score S_t,
- coherence-failure indicator $C_t \in \{0, 1\}$,
- safety severity $g_t \in \{\mathsf{PASS}, \mathsf{DELAY}, \mathsf{BLOCK}\}$ with mapping $\lambda(\mathsf{PASS}) = 0, \lambda(\mathsf{DELAY}) = 1, \lambda(\mathsf{BLOCK}) = 2$.

Aggregate:

$$\Delta P_w := S_{end} - S_{start}, \quad \Delta C_w := \sum_{t \in w} C_t, \quad \Delta \Lambda_w := \max_{t \in w} \lambda(g_t).$$

5.7 Public proxy energy and public gate

Define the computable proxy:

$$\Delta R^{pub}_{\mathrm{total}}(w) := \alpha \Delta P_w - \beta \Delta C_w - \gamma \Delta \Lambda_w,$$

with fixed public constants (default) $\alpha = \beta = \gamma = 1$. Define the public gate:

$$\mathrm{Gate}_{pub} : \ \Delta R^{pub}_{\mathrm{total}}(w) > \theta_{pub},$$

with fixed public threshold (default) $\theta_{pub} = 0$.

We set $\theta_{\mathrm{pub}} = 0$ as a neutral baseline for discussion: the gate fires only when progress outweighs coherence and safety penalties under the declared proxy.

5.8 Falsification (proxy-level, but now structurally bound)

Because Gate_{pub} is a necessary precondition of Gate_{evo}, the following is falsifiable:

Falsified if: any run claims an evolution activation while Gate_{pub} is false for the corresponding window w under the declared $(H, K, \alpha, \beta, \gamma, \theta_{pub})$.

Additionally (quality check), over N windows (default $N = 10$), flag failure if:

- (no-progress firing) Gate_{pub} fires in $\geq m$ windows (default $m = 5$) while median $\Delta P_w \leq 0$ on firing windows, or
- (block on firing) there exists a firing window with $\Delta \Lambda_w = 2$.

5.9 Minimal reporting template (one paragraph)

Report $(H, K, \alpha, \beta, \gamma, \theta_{pub}, N, m)$ and the observed series $\{\Delta P_w, \Delta C_w, \Delta\Lambda_w\}_{w=1}^{N}$ (or sufficient summaries to verify falsification).

5.10 Series mapping (one-box bridge for V4–V7)

- V4: action selection under value (EV) — "what to do"
- V5: Guard/SiriusA — "what must not be violated"
- V6: PIC — "how updates merge without rupture"
- V7: AGI door — "when self-recursion is structurally admissible"

This addendum specifies the missing bridge: PIC-style confluence is a necessary substrate for stable self-recursion, and the public gate is a necessary precondition for any evolution activation claim.

5.11 Witnessed predicates for A(S), G(S), and I(S) (minimal, non-operational)

To avoid vacuous predicates, we define A(S), G(S), and I(S) as witnessed predicates: they are considered satisfied only if accompanied by a publicly checkable witness artifact.

Aspire predicate $A(S)$. $A(S)$ holds if there exists a declared public harness H and a scalar aspiration witness $J_H(S) \in \mathbb{R}$ such that over a fixed windowing scheme (K, N):

$$\text{Median}_{w \in \mathcal{W}_{fire}}\big(\Delta J_H(w)\big) > 0, \quad \text{and} \quad J_H \text{ is non-trivial (not constant) on the run.}$$

Here $\mathcal{W}_{fire}$ denotes windows where Gate_{pub} fires. This does not disclose private aspiration internals; it only requires a public witness that distinguishes directionality from an "empty coherence loop."

Guard predicate $G(S)$. $G(S)$ holds if there exists a declared public severity mapping $g_t \in \{\mathsf{PASS}, \mathsf{DELAY}, \mathsf{BLOCK}\}$ and conservative aggregation (e.g., max on severity) such that:

$$\forall w \in \mathcal{W}_{fire},\ \Delta\Lambda_w \neq 2 \quad \text{(i.e., no } \mathsf{BLOCK} \text{ on firing windows).}$$

This again does not disclose private policy details; it only requires a public witness showing that evolution claims do not pass through BLOCK.

Intelligence predicate I(S) (witnessed, minimal). I(S) holds if there exists a declared public harness H' and a reproducible improvement trace over a fixed evaluation windowing scheme (K, N): (i) determinism under As-of: identical inputs (asof, delta, H', K) produce identical outputs (including the same canonical state hash), and (ii) non-trivial progress under the same H':

$$\text{Median}_{w \in W_{H'}} \Delta P_w > 0.$$

This defines I(S) as a witnessed, externally checkable predicate without claiming benchmark supremacy.

Thus AGI(S) holds when the three witnessed predicates A(S), G(S), and I(S) are simultaneously satisfied under their respective declared harnesses. Aspire specifies directional preference, whereas Intelligence specifies the capacity to realize reproducible improvement under that preference without rupture.

5.12 Related work pointers (non-exhaustive)

Our proposal is not a benchmark or universal intelligence measure; it is a structural necessary-conditions lens for long-run self-recursive operation. We point to: Legg & Hutter (universal intelligence measure), Chollet (benchmarking and generalization), Russell (alignment framing), Omohundro (basic AI drives), Amodei et al. (concrete safety problems), Shapiro et al. (CRDT/algebraic confluence), Turner et al. (impact/AUP-style constraints), and MIRI-style formal concerns on self-improvement. The novelty claim is the explicit bridge: confluence/canonicalization as a necessary substrate for stable self-recursion under conservative safety aggregation.

5.13 Commitment to the private gate (0.7, non-revealing)

To reduce arbitrariness while preserving the 0.7 boundary, we commit to a private-gate specification by publishing a versioned commitment hash $h = \mathrm{SHA256}(\texttt{spec_priv})$ alongside the release artifacts. This makes post-hoc moving of Gate_{priv} detectable without revealing its content.

The commitment hash is published at release time together with the version tag/DOI artifacts, so post-hoc changes are detectable.

5.14 Comparative positioning (relative strengths and limitations)

This work competes primarily on *structural definition* rather than capability benchmarks: we propose a necessary-conditions lens for long-run self-recursive operation, using (1) conjunction-style predicates $A(S) \wedge G(S) \wedge I(S)$, (2) a stabilization operator $F(s, o) = \mathrm{Canon}(s \sqcup \delta(s, o))$ grounded in confluence, and (3) an evolution activation gate with a public handle Gate_{pub}.

Axis E: subtractive perspective. Much of contemporary AI research emphasizes additive capability expansion, introducing new modules, parameters, or training objectives. In contrast, the present proposal begins from a subtractive perspective: identifying minimal structural constraints required to prevent rupture in long-run self-recursive operation.

Axis A: definition stability under benchmark drift. Relative to benchmark-centered AGI discussions, a structural necessary-conditions definition is less sensitive to benchmark leakage/inflation. Relative to fully formal intelligence measures, this approach trades formal sharpness for operational robustness.

Axis B: self-recursion under stable integration. By tying self-update stability to confluence/canonicalization, the proposal directly addresses order-dependence and last-write-wins failure modes in long-run operation. However, the public portion remains a structural proposal; full theorem/proof development is outside the disclosed scope.

Axis C: safety constraints as non-negotiable components. Guard is treated as a constitutive constraint rather than an add-on. The conservative aggregation perspective (PASS<DELAY<BLOCK with max-style merging) makes the safety stance non-dilutive under merges, but detailed policy calibration remains private (0.7), limiting third-party comparability.

Axis D: falsifiability and reproducibility. To avoid "proxy substitutes the core" concerns, we define $\mathrm{Gate}_{evo} := \mathrm{Gate}_{pub} \wedge \mathrm{Gate}_{priv}$, so the public gate is a necessary precondition for any claimed

evolution activation. The public handle ($\Delta P_w, \Delta C_w, \Delta\Lambda_w$, fixed window K) enables proxy-level falsification reports, while private calibration is intentionally withheld.

This framing treats stability constraints as primary, and views additional capability as secondary to maintaining non-rupturing integration over time.

5.15 Formal bridge: Why Aspire is necessary beyond PIC

PIC (A1–A3) guarantees order-robust, idempotent, and monotone integration over the state space S. However, a join-semilattice structure is direction-neutral: it preserves confluence but does not induce directional preference.

Formally, under A1–A3 alone, repeated merge and canonicalization converge to stable fixed points, but no term in the algebra selects among equivalent consistent trajectories.

Aspire introduces a directional witness J_H that imposes a monotone preference ordering over S. Thus A(S) is not an arbitrary addition but a necessary augmentation to PIC for non-trivial self-recursive evolution.

Minimal formalization of Aspire Let $J_H : S \to \mathbb{R}$ be a public aspiration witness monotone over the PIC order. If $s \preceq s'$ (i.e., $s \sqcup s' = s'$), then $J_H(s) \leq J_H(s')$.

A(S) holds when J_H is non-trivial and strictly increases across firing windows.

In practice, such witnesses may arise from iterative counter-model comparisons, multi-agent reasoning loops, or other mechanisms that accumulate directional evidence across turns.

5.16 Minimal MMAR log example (CI artifact)

We provide a minimal empirical trace showing that MMAR can detect and persist candidate structural signals under CI.

CI run. GitHub Actions (MMAR Recurrence v0) produced `.mmar/recurrence_log.json` and uploaded it as an artifact.

Excerpt (recurrence_log.json).


```
{
  "11832dd82f764556": {
    "type": "NOVEL_STRUCTURE",
    "first_seen": "2026-03-04T23:37:00.641810+00:00",
    "count": 1,
    "sources": [{"case_id":"demo-001","asof":"2026-01-23"}],
    "support": ["type:NOVEL_STRUCTURE","signal_key:block"]
  }
}
```


Artifact commitment. The artifact zip was uploaded with SHA256:
`fbdcde1f7be2199a2d1ccb6f56a5d303e91c11c9fd96463e50c458d4d3078345`.

Mapping MMAR recurrence types to public observables. We map recurrence signals to the public observables as follows (minimal convention): NOVEL_STRUCTURE contributes to ΔP_w (candidate progress count); DISAGREEMENT and STRUCTURAL_ANOMALY contribute to ΔC_w (coherence-cost events); MISSING_EVIDENCE contributes to $\Delta \Lambda_w$ (safety-risk proxy, typically DELAY unless explicitly declared as BLOCK).